

Scalable Moment-Based Filtering and Extensions to Hybrid Systems

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Abstract

A central obstacle in nonlinear Bayesian filtering is representing the belief distribution. Moment-based filters address this by propagating polynomial moments and reconstructing a density from them. Recent work completes the predict-update loop via the maximum-entropy (MaxEnt) principle, but each step requires the partition function and its gradient, both n -dimensional integrals whose cost scales exponentially, restricting the demonstrated MaxEnt moment filtering to $n \leq 4$. We avoid the partition function entirely by combining score matching with Stein's identity. In our setting, score matching reduces the density fit to a single linear solve whose coefficients are assembled directly from the propagated moments. The same parameters then drive Stein's identity to close the moment hierarchy during prediction and to recover posterior moments after each Bayesian update, keeping the full predict-update loop free of partition function evaluation. The resulting Score Kalman Filter (SKF) reduces to the classical information-form Kalman filter as a special case and performs every step through linear algebra. On nonlinear coupled-oscillator networks, the SKF runs through $n = 20$ and reports lower RMSE than the EKF, UKF, EnKF, and particle-filter baselines on the tested synthetic benchmarks. In this talk, I will also discuss extensions to hybrid systems and future research efforts to extend the mathematical framework to jet bundles for field estimation.

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